

ASCII MASTER FLOW – PANEL 1/12: Greenhouse-effect firewall

NATURAL EFFECT -> absorbs and re-emits outgoing longwave radiation

FUNCTION -> maintains a habitable energy balance

HUMAN ENHANCEMENT -> raises concentrations and net forcing

CLIMATE RESPONSE -> warming plus wider system changes

EXAM RULE -> preserve the natural-versus-enhanced distinction

MUST REMEMBER: Climate change follows radiative forcing and Earth-system feedbacks from...

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ASCII MASTER FLOW – PANEL 2/12: Energy-balance rail

SUN -> incoming predominantly shortwave energy

SURFACE -> absorbs part and warms

EARTH -> emits outgoing longwave infrared energy

GREENHOUSE GASES -> absorb and re-emit part of longwave flow

IMBALANCE -> forcing changes energy retained by the climate system

ASCII MASTER FLOW – PANEL 3/12: Emission-to-concentration ledger

ACTIVITY -> emissions flow during a stated period

ATMOSPHERE -> concentration after sources and sinks interact

ACCUMULATION -> long-lived gases build an atmospheric stock

FLOW CUT -> slows addition but does not erase the stock

REMOVAL -> changes the stock only within a stated boundary

ASCII MASTER FLOW – PANEL 4/12: Forcing-feedback fork

EXTERNAL OR IMPOSED ENERGY CHANGE -> radiative forcing

CLIMATE RESPONSE TRIGGERED -> feedback

AMPLIFIES INITIAL CHANGE -> positive feedback

DAMPENS INITIAL CHANGE -> negative feedback

SIGN -> direction of response, never moral value

ASCII MASTER FLOW – PANEL 5/12: Water and particle matrix

WATER VAPOUR -> greenhouse gas commonly acting as feedback

SULPHATE-LIKE AEROSOLS -> often cooling influence

BLACK CARBON -> absorbing warming influence

SNOW OR ICE DEPOSITION -> can lower albedo

BOUNDARY -> agent, location and mechanism determine the sign

ASCII MASTER FLOW – PANEL 6/12: Gas comparison gate

GWP -> relative integrated influence over a named horizon
POTENCY -> effect per unit mass under the metric
LIFETIME -> persistence in the atmosphere
EMISSION SCALE -> quantity released
POLICY -> near-term and cumulative levers are not identical

ASCII MASTER FLOW – PANEL 7/12: Weather-to-climate ladder

EVENT -> individual weather occurrence

SEASON OR YEAR -> short-period variability

DISTRIBUTION -> frequency, intensity and pattern

LONGER RECORD -> climate trend assessment

RULE -> fluctuation around a trend is not trend reversal

ASCII MASTER FLOW – PANEL 8/12: Attribution evidence chain

OBSERVATION -> define the measured change

DETECTION -> distinguish signal from expected variability

DRIVERS -> compare human and natural influences

MODELS PLUS PHYSICS PLUS RECORDS -> converging evidence

ATTRIBUTION -> qualified contribution, not slogan causation

ASCII MASTER FLOW – PANEL 9/12: Event-claim firewall

DEFINE EVENT CLASS -> place, duration and metric
COMPARE WORLDS -> with and without human influence
ESTIMATE CHANGE -> probability or intensity
RETAIN OTHER DRIVERS -> exposure, vulnerability and variability
NO SOLE CAUSE -> disaster impact is multi-causal

ASCII MASTER FLOW – PANEL 10/12: Observation-projection matrix

OBSERVATION -> measured past or present

ATTRIBUTION -> explanation of observed change

SCENARIO -> conditional assumptions

PROJECTION -> modelled response under a scenario

FORECAST -> different claim with a prediction horizon

CLOSE DISTINCTION: Weather is not climate, emission flow is not concentration stock,...

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ASCII MASTER FLOW – PANEL 11/12: Climate-response fork

SOURCE REDUCTION -> mitigation

SINK ENHANCEMENT -> mitigation with accounting boundary

EXPOSURE OR VULNERABILITY REDUCTION -> adaptation

RESIDUAL HARM -> separate loss-and-damage question

PORTFOLIO -> mitigation and adaptation are complementary

EVIDENCE LIMIT: MECHANISM / STATUS / CAUSATION: State unit, baseline, period, scenario...

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ASCII MASTER FLOW – PANEL 12/12: Science answer spine

DEFINE -> natural effect, human enhancement and energy flow

ACCOUNT -> emissions, concentrations, stock and net balance

EXPLAIN -> forcing, feedback and gas-specific properties

EVALUATE -> trend, detection, attribution and projection

QUALIFY -> scale, scenario, confidence and no invented figures